

Using Bayesian Network Inference and underwater imagery to understand the influence of environmental heterogeneities on benthic community structure in the Antarctic Peninsula

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Author contributions

LK, EM and BD conceived and developed the project. All data was collected by LK and BD during the TANGO1 expedition. LK analysed the data, with the assistance of TMK and EM. LK wrote the first draft, and all authors contributed to editing the paper and the final version.

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Abstract

The Western Antarctic Peninsula (WAP) is one of Earth's fastest-warming regions. Increasing temperatures and winds are disrupting sea-ice dynamics, which can impact benthic communities. Local heterogeneity in these benthic communities enhances biodiversity and provides redundancy, critical for ecosystem resilience. Therefore, understanding local heterogeneity and its primary drivers is essential to understand how benthic communities may

shift under climate change. To assess local heterogeneity, we combined underwater imagery with a Bayesian network approach. Video sampling was done using a portable Remotely Operated Vehicle across four sites in Dodman Island, a small bay (1 km²) within the WAP. The dataset comprised 0.8 m² photographs taken within 10 m² areas, located within 50 m from each other to form a site. All data was collected within one week during Antarctic summer. We annotated seafloor images for morphotaxa and substrate type, then used Bayesian Network Inference (BNI) to infer an ecological network of dependencies between taxa and key environmental variables. Our results indicate high small-scale heterogeneity in benthic communities between sites, both in taxonomical and functional groups. In our Bayesian network, substrate granulometry, maximum depth, and distance to the glacier were the most connected environmental variables; only starfish showed similar connectivity among taxa. Using BNI, we inferred how changes propagate through the network via direct or cascading interactions. Substrate granulometry had a strong impact on all taxa, while macroalgal abundance changes had relatively weak impact. We demonstrate how BNI, coupled with non-destructive sampling, can support predictions of benthic change and inform conservation priorities.

Keywords: Bayesian networks, underwater imagery, benthos, Antarctic peninsula, climate change

Introduction

The Western Antarctic Peninsula (WAP) is one of the fastest warming areas on Earth. This fast warming is due in part to the Southern Annular Mode, the leading mode of atmospheric circulation in the Southern Ocean, which is influenced by ozone depletion, increased winds and global warming, contributing to its positive trend (Mindlin et al. 2021). The impact of this positive phase in the Antarctic Peninsula has led to disruptions in sea-ice dynamics, including alterations in the duration of the sea-ice season, extended glacier retreats, ice shelf collapses, and increased iceberg scouring (Turner et al. 2014; Pörtner et al. 2019; Siegert et al. 2019; Eayrs et al. 2021). These disruptions to Antarctic benthic habitats, which host most of the region's biodiversity, could drive changes which will trigger cascading effects throughout the ecosystem (Rossi et al. 2019; Caputi et al. 2020) since sea-ice shapes benthic community composition and function (Cummings et al. 2006; Clark et al. 2017; Gutt et al. 2019). A recent study by Amsler et al. (2023) around the WAP has, for example, shown the negative correlation between sea-ice cover and macroalgal cover. Additional studies in the WAP have shown that glacier retreat affects macrobenthic composition, with macroalgae patches colonizing the newly ice-free zones (Pasotti et al. 2015), and that sedimentation causes shifts from ascidian-dominated communities to mixed assemblages (Sahade et al. 2015).

The WAP is a hotspot for macrobenthos (Gutt et al. 2016). However, small-scale patchiness within this region is often overlooked in broader studies (Gutt et al. 2019). This local heterogeneity likely provides redundancy that is important for ecosystem resilience (Hewitt 2005) and enhances local biodiversity (Romoth et al. 2023). Understanding these patterns in heterogeneities and identifying the primary drivers behind them is crucial for a comprehensive understanding of the WAP's benthic communities. By examining how these drivers may shift under climate change, we can better predict potential impacts on marine ecosystems. Studying climate change's effects on marine communities is however complex due to the interaction of numerous factors, both biotic (species interactions) and abiotic (environmental characteristics). Therefore, a model that incorporates both is essential for a holistic view of the processes shaping benthic communities in the WAP.

Bayesian Networks have been increasingly used as a tool to study community structure and ecological relationships around the world (Forio et al. 2023) and in the Southern Ocean (Mitchell et al. 2020; Khan et al. 2024). They are probabilistic models that infer dependencies between variables, without making assumptions about the nature of the variable or the relationship (Heckerman 1995). This approach allows for more flexibility than other models since it doesn't assume linearity in parameters and the model is learned from the data itself rather than having to be specified beforehand (Heckerman 1995; Friedman et al. 1998; Forio et al. 2023). Bayesian Networks are thus becoming increasingly popular for studying habitat suitability, modelling complex environmental systems, and scenario forecasting (Milns et al. 2010; Pozsgai et al. 2016; Sander et al. 2017; Mitchell et al. 2020, 2021; Hui et al. 2022). They are highly relevant to use for understanding the influence of environmental factors on benthic systems (Mitchell et al. 2020), and enabling the inference of past and future ecosystem states (Mitchell and Harris 2020; Mitchell et al. 2021).

Given the increasing intensity and rapid pace of global change, there is an urgent need to establish a baseline of benthic biodiversity, outline the scale at which there is heterogeneity in the Antarctic, and develop novel conservation strategies that consider multiscale variability and patchiness (Kennicutt et al. 2015; Gutt et al. 2019). Achieving this baseline requires the development of standardized, cost-effective, repeatable, and easy-to-implement field protocols for comprehensive and consistent data collection. Additionally, efficient data treatment workflows, novel statistical approaches, and Open Access to all collected data are essential to accelerate research, as the current pace of scientific progress is likely insufficient to keep up with the rapid climate changes (Roche et al. 2022). Underwater imaging techniques provide a robust method for *in situ*, non-destructive, quantitative, and repeatable sampling of benthic communities (Durden et al. 2016). This approach has been validated in numerous prior studies (Segelken-Voigt et al. 2016; Brasier et al. 2018; Gutt et al. 2019; La Mesa et al. 2022; Khan et al. 2024). Using a small Remotely Operated Vehicle (ROV) for image acquisition can overcome the challenges associated with diving in extreme environments, thereby enabling surveys to reach greater depths, cover more extensive areas, and significantly enhance sampling time efficiency, which is usually very expensive in remote areas such as the Antarctic (Piazza et al. 2020; Amsler et al. 2023).

In this study, using a non-destructive, easily replicable method based on ROV imagery, we surveyed benthic communities of the WAP to document small-scale (1 km²) heterogeneity. We identified the major environmental parameters driving these community differences using Bayesian Network inference and used that model as a base for inferring community changes associated with climate change.

Material & Methods

Sampling station

The sampling took place during the TANGO1 expedition between February and March 2023 aboard a nimble sampling platform, the RV Australis, a steel-hulled, fully rigged motor sailor (Danis et al. 2023, 2024). The study was conducted at Dodman Island (66.00 °S, 65.78 °W) in the Grandidier Channel. This protected inner bay was selected for its small size (1 km²) and its environmental settings, particularly the gradients of exposure to ice-related factors such as annual sea-ice cover and glacier activity. The bay's seabed is characterized by muddy sediments interspersed with gravels and dropstones, creating a highly variable distribution of benthic habitats within a small area. Four sites were chosen along the bay's natural gradients, each with contrasting geomorphological and physical characteristics, including proximity to the active glacier, slope, and substrate type (Figure 1). Sites 1, 2 and 3 are pool-like sites. Sites 1 and 3 are located directly under the active glacier. Site 2 occupies an intermediate position (at ca. 400 meters from the glacier). Site 4, the furthest from the glacier (ca. 600 meters), is open to the Mudge passage, which goes to more than 300 meters depth. All sites are within the same depth range of 15-28 meters.

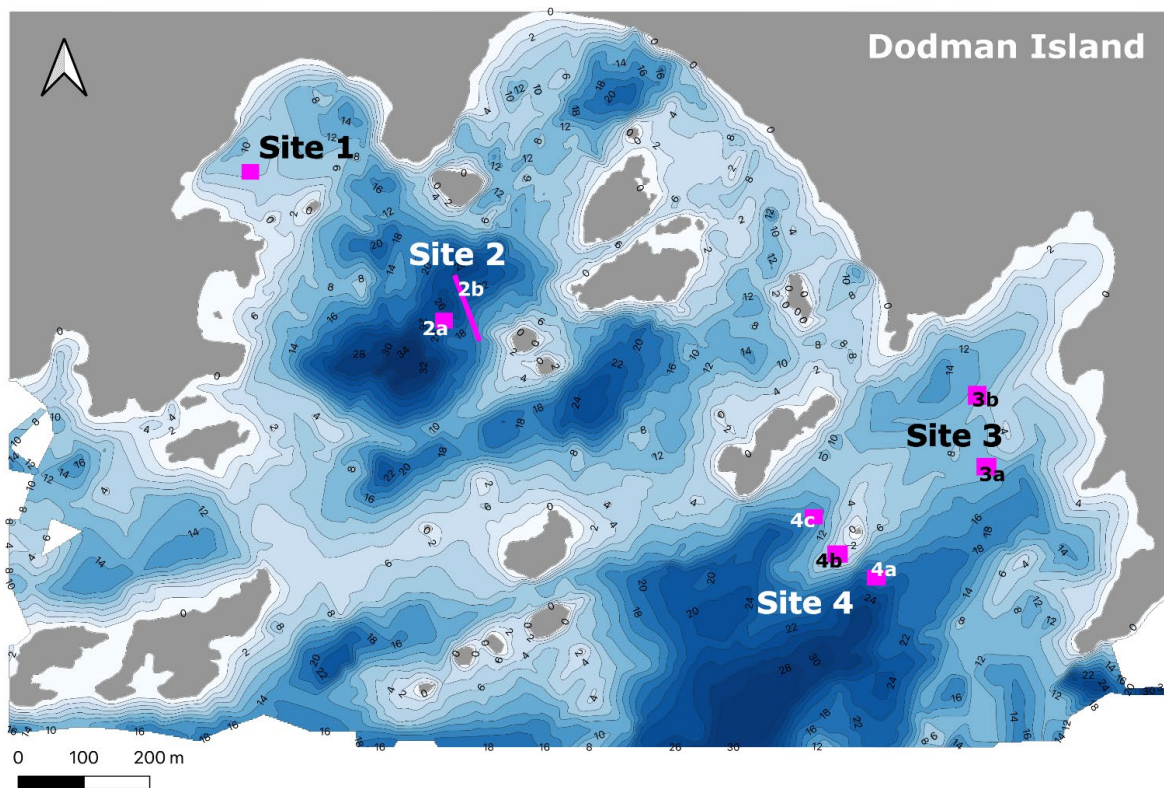


Figure 1 – Location of the 4 different sampling sites in Dodman Island (66.00 °S, 65.78 °W). In pink, the exact location of the replicates for each site.

Image acquisition

Benthic image sampling was conducted using a small, portable ROV (BlueROV2 Heavy, BlueRobotics) equipped with four 1500-lumen lights, a built-in front-looking camera, and a down-looking GoPro HERO 10 camera, which captured detailed images of the seafloor. Additionally, synchronized temperature, heading, depth, bathymetry (using a single-beam sonar), and real-time positioning (with an acoustic GPS, WaterLinked) were recorded. At each sampling site, the ROV was flown at a constant speed (<1 m/s, as slow as possible to maintain high image quality) and altitude (approximately 1 meter from the bottom, adjusted for visibility), covering areas of ~ 10 m² using both “lawn-mower” trajectories or line transects. Three videos were recorded at each site, with a maximum of 50 meters between areas within the same site and around 250 meters between each site. Therefore, these videos provided spatial resolution at the meter scale (photographs), ten-meter scales (areas), and 100-meter scale (sites). All videos of Dodman Island were recorded within the span of a week.

Image analysis

Non-overlapping photos were extracted from the videos using FFmpeg (version 6.0.1, Bellard 2023). To correct and homogenize the color and illumination, these photos were modified using the Matisse pre-processing tool (Ifremer, Arnaubec et al., 2015). This extra step allowed for a better success rate in the next stages of our workflow.

Annotations were conducted in BIIGLE (version 2.0, Langenkämper et al. 2017) using a custom label tree based on the CATAMI annotation scheme (Althaus et al. 2015), for both taxa and substrate features. Given the challenges of taxonomic identification from photos without physical samples (Althaus et al. 2015; Horton et al. 2021), organisms were labelled based on distinct morphological features, with as much detail as possible (Khan et al. 2024). The CATAMI scheme offers a consistent and standardized vocabulary, adaptable to various spatial scales and image qualities (Althaus et al. 2015). To provide a more detailed identification of the Antarctic benthos, additional levels were incorporated into the base structure of the CATAMI label tree (Katz 2024) while maintaining the original standard format. The assigned identification is the finest taxonomic resolution that is robust based on photo samples only.

Since the identification level varied among taxa (ranging from species to genus or family), we use the term "morphotaxa" to refer to the identified biota throughout this study. To further hone our taxonomic identification, we were able to confidently assign species names to several morphotaxa (Table 1) as some specimens of these taxa were hand-picked by SCUBA divers (Danis et al. 2023).

For each photo, the position, size, and shape of the individuals/colonies, along with the substrate type and the presence of dropstones, were recorded. The cover of each annotation was calculated from the size ratio to the total photo area ($\sim 0.8 \text{ m}^2$). To improve accuracy, calibration of the down-looking camera was performed by taking a series of photographs that include a scale bar using the same settings. The resulting size-to-pixel ratio allowed us to estimate the total size of photos and organisms. In total, we obtained a dataset of 297 annotated photos (28 in site 1, 89 in site 2, 31 in site 3 and 149 in site 4) which included 1179 annotated individuals (or colonies), 37 different identified morphotaxa (Table 1) and 8 different substrate types (Table 2).

Morphotaxa/BIIGLE annotation	description/ID	Trophic group	CATAMI code	CATAMI label	BNI node
beige_bivalve_msp1	<i>(Aequiyoldia eightsii)</i>	mix	23199000	Molluscs: bivalves	bivalves
clam_msp2		suspension feeders			
brown_bivalve_msp3	<i>Laternula elliptica</i>	mix	24000000	Molluscs: Gastropods	laternula
laternula		suspension feeders			
limpet	<i>Nacella concinna</i>	grazer	24000000	Molluscs: Gastropods	limpets
ascidian_msp1	<i>(Cnemidocarpa verrucosa)</i>	suspension feeders	35000902	Ascidians: Unstalked: Solitary	suspension feeders
ascidian_msp2					
ascidian_msp3	bright yellow				
	white				
yellow anemone msp2	long tentacles, yellow	suspension feeders	11229903	Cnidaria: True anemones: Other anemones	suspension feeders
finger_sponge_msp1	<i>Homaxinella balfourensis</i>	suspension feeders	10000915	Sponges: Erect forms: Branching	
yellow_blob_msp2			10000904	Sponges: Massive forms: Simple	
orange_blob_msp5			10000916	Sponges: Erect forms: Simple	
fine_orange_msp4			10000905	Sponges: Massive forms: Balls	
spiky ball sponge msp6					
desmarestia_msp1	<i>Desmarestia antarctica</i> <i>Desmarestia anceps</i> <i>(Plocamium hookeri)</i> <i>(Trematocarpus antarcticus)</i> <i>Himantothallus grandifolius</i> <i>Iridaea cordata</i>	photosynthesis	80300908	Macroalgae: Erect fine branching: Brown	macroalgae
desmarestia_msp2			80300904	Macroalgae: Erect coarse branching: Brown	
branching_red_algae_msp1			80300910	Macroalgae: Erect fine branching: Red	
branching_red_algae_msp2			80300902	Macroalgae: Large canopy-forming: Brown	
himantothallus_msp1			80300925	Macroalgae: Sheet-like / membraneous: Red	
red_sheet_msp1			80300922	Macroalgae: Sheet-like / membraneous	
red_or_brown_sheet_algae			80300000	Macroalgae	
macroalgae			80300934	Macroalgae: Encrusting: Red: Calcareous	encrusting algae
pink_encr_algae	<i>(Leptophytum coulmanicum)</i>		80300928	Macroalgae: Encrusting: Green	
green_encr_algae					
pink_star_msp1	<i>Odontaster validus</i> very small, white chunky arms, orange	mix	25102000	Echinoderms: Sea stars	starfish
glitter_star_msp3					
carrot_star_msp4					
brown_star_msp9					
skinny_yellow_star_msp10					
medium_yellow_star_msp13					
light_brown_star_msp15					
asteroidea					
urchin_msp1	<i>Sterechinus neumayeri</i>	mix	25200901	Echinoderms: Sea urchins: Regular urchins	urchins
parborlasia_corrugatus	<i>Parborlasia corrugatus</i> pile of multiple <i>Parborlasia</i> other than <i>Parborlasia</i>	mix	80600901	Worms	worms
parborlasia_aggr		mix			
Worms		mix			
Echiura		deposit feeder			

Table 2 – Identified substrate categories.

BIIGLE annotation	Group	CATAMI code	BNI node	node category
Substrate invisible	/	NA	substrate granulometry	NA
Fine sand (no shell fragments)	Unconsolidated	82001015		Fine sand
Coarse sand (with shell fragments)		82001014		Coarse sand
Gravel (2-10 mm)		82001011		Gravel/pebbles
Pebble (10-64 mm)		82001012		
Cobbles	Consolidated	82001004		Boulder/rock
Boulder		82001003		
Rock		82001002		

Statistical analysis

Biodiversity

Biodiversity indices were computed using the “vegan” R package (version 2.6, Oksanen J et al. 2022), in R version 4.3.1 (R Core Team 2023). Alpha diversity was assessed using the Shannon-Wiener index, with higher values indicating greater diversity. Pielou’s evenness index was calculated to measure the distribution of morphotaxa within the community and is a measure of dominance. It ranges from 0 to 1, with a low evenness meaning that one or few morphotaxa dominate the community.

Bayesian Network Inference analysis

Bayesian Network Inference (BNI) represents ecosystems as a network, with variables being treated as “nodes” and dependencies between them as “edges”. Here, the nodes are morphotaxa and physical parameters (substrate granulometry, distance to the glacier, slope and maximum depth). These dependencies are found based on frequent co-localizations (or lack thereof).

The Banjo algorithm uses uniform priors, which requires discretisation of the count data into equal bins (Heckerman 1995; Yu et al. 2002, 2004; Milns et al. 2010). This discretisation balances out overfitting to noisy data with the power to detect the conditional dependencies between two nodes (Milns et al. 2010). By considering the relative abundance of each taxonomic group, the process ensures that very high abundances do not dominate all signals, and rarer groups do not get masked. Furthermore, this approach treats all taxa and environmental variables as equally important at the input level. The resulting dependencies can be positive, i.e. Node A depends on Node B, negative, Node A depends inversely on Node B as well as non-monotonic, i.e. different behaviour at different Node values. These non-monotonic dependencies capture density-dependent behaviour. Abundance data was transformed from counts into discrete bins

(binomial: presence/absence), which has been shown to work well for ecological data (Milns et al. 2010). Similarly, environmental parameters were categorized into discrete levels (Brasier et al. 2018): substrate type (*substrate granulometry*) was classified into five categories from soft to hard (fine sand, coarse sand, gravel/pebbles, boulders/rock, Table 2), *distance to the glacier* was categorized into three levels (adjacent, close, further), *slope* was defined by the difference between minimum and maximum depth of the sampled photos and was classified into 2 levels (flat or mild slope) and *maximum depth* was classified into 5 different ranges (0-5m, 6-12m, 13-18m, 19-25m, 26-32m).

To avoid zero-inflated data (much more “absence” than “presence”, for rarer taxa) and so avoid Type I errors, because the priors of the BNI are uniform distribution (Milns et al. 2010), many morphotaxa were grouped by functional traits (Suppl. Figure 1). In the end, 13 nodes were kept: *laternula*, *urchins*, *starfish*, *limpets*, *macroalgae*, *encrusting algae*, *suspension feeders*, *substrate granulometry*, *dropstones*, *distance to the glacier*, *maximum depth*, *slope*, and *site* (Table 1, Suppl. Figure 2). In contrast, for prevalent biotic or abiotic items (much more “presence” than “absence”), additional levels could be utilised. *Macroalgae* and *dropstones* were discretized in this way, where abundances at or below the median were categorized as “few,” while values above the median were categorized as “many” (cf. (Khan et al. 2024)). All data preparation was done in R version 4.3.1 (R Core Team 2023).

To infer the dependencies (i.e. the connections) between taxa and environmental parameters (nodes) a search was performed using Banjo (Heckerman 1995), a Java-based software. Using the data that was discretized in the previous steps, Banjo generates a random network and then performs 10 million searches by adding and subtracting edges for a more likely network than the random one that was generated. The final output is the most probable network with the strength of the dependency given by the Influence Score, which can be positive (aggregation between two nodes) or negative (segregation), and how strong this association is, with 1 indicating perfect association and -1 the inverse. To minimize outlier bias, we bootstrapped 1000 samples at a 95% level (cf. Mitchell et al. 2020) and the final network was defined as the edges that occurred in more than 60% of the analyzed networks (Mitchell et al. 2020; Khan et al. 2024). The direction of the dependencies was also assessed: if the edge frequencies from A to B and from B to A were around 50% ($\pm 10\%$), the dependency was considered bi-directional (cf. Mitchell et al. 2020). In our final graph, we represented our edges as arrows, with their colour indicating the type of connection (blue = abiotic to abiotic, orange = biotic to biotic, red = abiotic to biotic, yellow = biotic to abiotic), their arrowhead in the direction of the dependency (or no arrowhead if the dependency is bi-directional) and the width of the line as the probability (or frequency). Once we have our final network, we can calculate the probability of a certain node (A) being in a given state given another node (B) is in a given state, using the following formula:

$$P(A|B) = \sum_{n=1}^{n=N-1} \frac{\prod_{s=1}^{s=S} P(B_n|A_{n+1})}{\sum_{m=1}^N P(B|A_m) P(A_m)}$$

Where N is the total number of nodes in the chain connecting node A to B, n and m are the chain’s indices, S is the number of discrete states for each node and s is the index of the state. Using this formula, conditional probabilities

were estimated for all nodes using the “bnlearn” R package (version 5.0, Marco Scutari 2010) and then summarized for each biotic node. We used this probability table to infer the impact of changes in biotic nodes’ presence/absence due to changes in environmental variables by calculating by using the difference between the probability of presence in one state of the node with the probability of presence in another state of that same node.

Results

Spatial heterogeneity

Due to the high particle content in the water column, some videos did not have sufficient quality to enable robust identification and analyses, so were removed along with subpar square replicates. After quality control, one replicate was kept for site 1, the nearest to the glacier. This comprises a mixture of soft bottoms and large dropstones. The area is predominantly inhabited by limpets (*Nacella concinna*) and their dwelling traces (Figure 3-a). Of the two Site 2 replicates, site 2a features a soft bottom with scattered algae patches. Dropstones in this area are colonized by ascidians (Figure 3-b). Site 2b exhibits a soft bottom with smaller dropstones and small algae patches, as well as coarse sand containing numerous bivalve shells. Both the Site 3 replicates were characterized by a soft bottom with a high density of the burrowing mollusc *Laternula elliptica* (>40 individuals/m²), along with a macroalgae forest predominantly composed of both species of *Desmarestia*. Additionally, multiple aggregates of the giant nemertean *Parborlasia corrugatus* were observed (Figure 3-c). Of the three site 4 replicates, sites 4a and 4b are dominated by macroalgae, with some patches of soft bottom interspersed while site 4c is characterized by rocky terrain with sheet-like algae dominance, along with the limpet *Nacella concinna*.

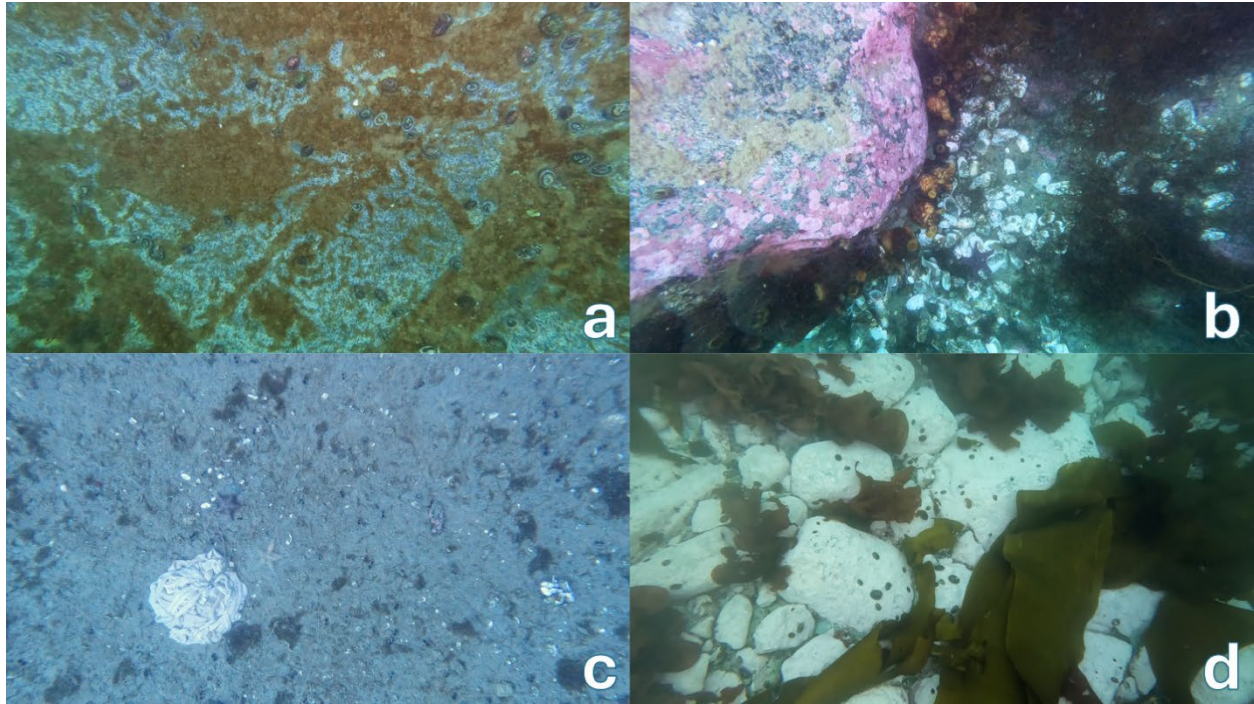


Figure 2 – Sample images from the ROV, taken at approx. 1 m from the bottom. a) site 1: limpets and dwelling traces on rock. b) site 2b: dropstone with pink encrusting algae and ascidians (msp1). c) site 3a: pile of *Parborlasia corrugatus*. d) site 4c: rocks with sheet-like macroalgae and limpets.

Diversity metrics were highly variable among our sites (Figure 4). The patterns across the sites in the number of distinct morphotaxa exhibit different patterns from the Shannon and Pielou indices: the number of distinct morphotaxa is the highest in sites 2b and 3a (22), and the lowest in site 1 (10). The Shannon diversity index is the lowest in site 3b (0.3386) and the highest in site 2a (1.8403). Pielou's evenness index patterns are similar to the Shannon Index patterns, meaning that the sites showing more alpha diversity also have the most even communities. Site 1 demonstrated a relatively high Pielou index (0.5838), indicating some evenness in the community, but had a lower number of morphotaxa. Sites 3a and 3b exhibited lower evenness indices (0.2032 and 0.1171, respectively) but a high number of morphotaxa (22 and 18 distinct morphotaxa), indicating an uneven community with one or more dominant morphotaxa. Site 4c showed a low alpha diversity (0.4615), with a low evenness index (0.1704).



Figure 3 - Biodiversity metrics of Dodman Island sites. **Distinct morphotaxa** (left) is the number of different morphotaxa in each site. The **Shannon index** (middle) is a measure of diversity, with a higher number meaning more diversity. The **Pielou index** (right) is a measure of evenness, where a low number means one or few taxa are dominant, and a higher number means a more even community. The colours indicate the different values, with blue representing the higher values, brown the middle values and yellow the lower values.

High spatial heterogeneity between sites was observed in the communities in terms of both taxonomic groups (Figure 5) and trophic groups (Figure 6). Site 2a and 2b exhibited a high proportion of tunicates (ascidians) compared to other sites. Sites 3a and 3b showed a high proportion of nemerteans, primarily aggregations of *Parborlasia corrugatus*. Echinoderms, with a dominance of the sea star *Odontaster validus*, were prevalent in sites 4a and 4b, while site 4c was characterized by molluscs, particularly limpets (*Nacella concinna*). From a trophic point of view, sites 1 and 4c were dominated by grazers, with a high abundance of limpets, while sites 2a and 2b were dominated by suspension feeders. Sites 3a, 3b, 4a, and 4b had a majority of animals with a mixed feeding strategy. Total macroalgae cover varied across sites, ranging from 1.26% in site 1 to 23.14% in site 2b. Macroalgae morphotypes also differed among sites (Figure 7), with either sheet-like algae or erect branching algae dominating, but not both. Encrusting algae were present in nearly every site (Figure 7).

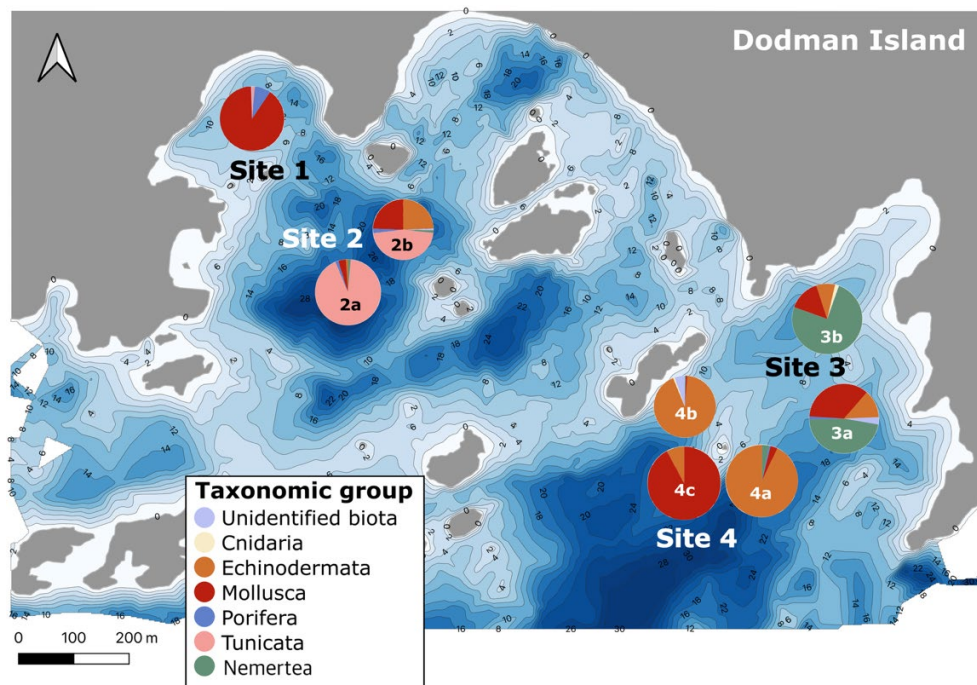


Figure 4 - Map of Dodman Island. Pie charts represent relative cover of taxonomic groups identified on the photos. Macroalgae were removed from the chart for clarity.

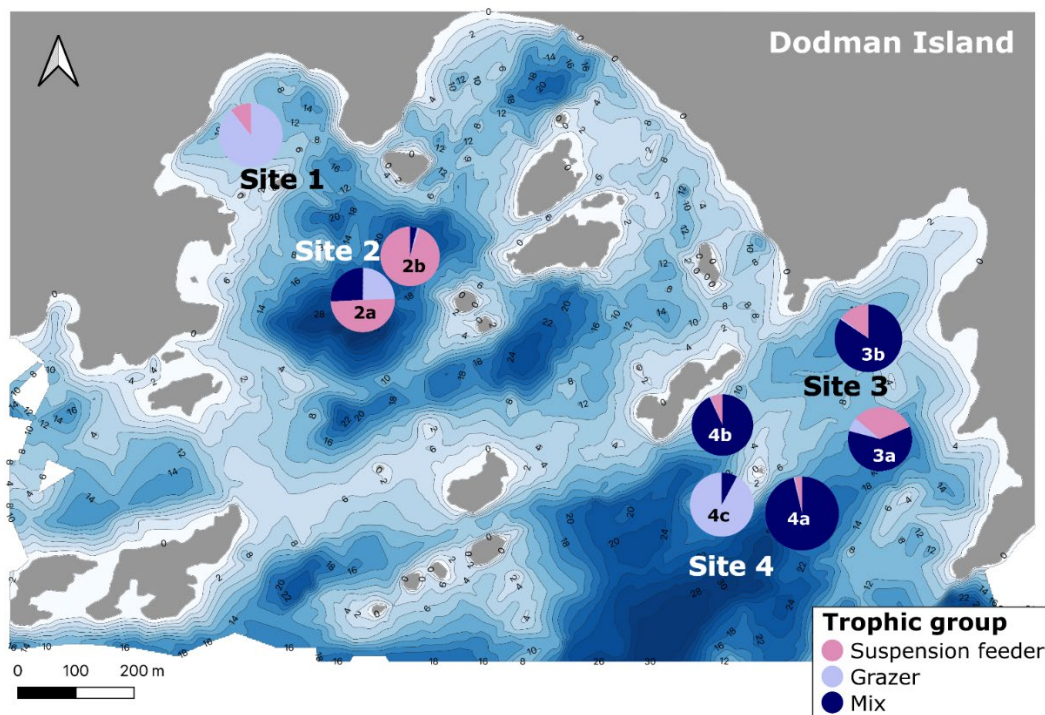


Figure 5 - Map of Dodman Island. Pie charts represent relative cover of trophic groups identified on the photos. Macroalgae were removed from the chart for clarity because they contributed between X and Y percent of each site.

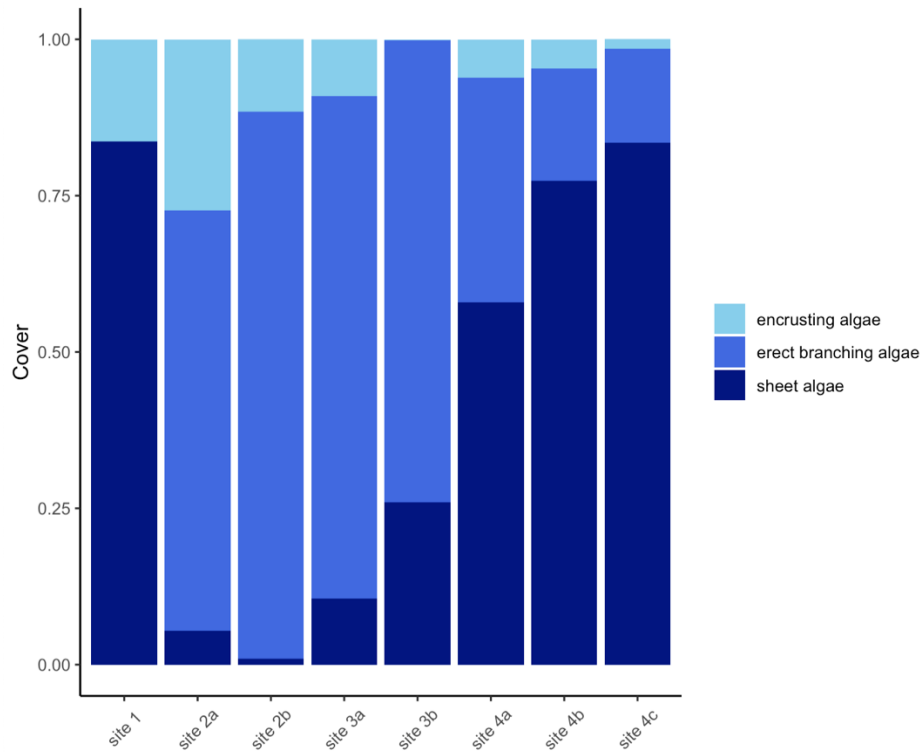


Figure 6 - Macroalgae relative cover in Dodman Island sites.

Ecological network

BNI analysis revealed 26 dependencies (or edges) between our nodes in the network (Figure 8). Between our abiotic nodes, *substrate granulometry*, *maximum depth*, and *distance to the glacier* had the most dependencies (five edges). *Starfish* was the most connected taxon with five edges. The rest of the nodes had either three or four edges. The strongest dependencies (highest influence score, IS) in the network were between *maximum depth/slope* (IS = 0.8104) and *dropstones/limpets* (IS = 0.6868). Our edges were characterized by 10 negative and 16 positive edges. Thirteen edges exhibited frequencies above 90%, indicating even stronger and more reliable dependencies within the inferred network structure.

Node probabilities

Conditional probabilities of each biotic node's presence/absence were calculated across all node states (Suppl. Table 2).

Abiotic nodes

Substrate granulometry: on the photos where the substrate isn't visible, there is a 100% probability that *macroalgae* are present. Fine sand substrate has a high probability (>75%) of the absence of *laternula*, *urchins*, *limpets*, *encrusting algae*, and *suspension feeders*. On the contrary, *starfish* have an 88% probability of being present in this

substrate category. Conditional probabilities of presence for *macroalgae* behave differently, with a lower probability of being present on “fine sand” (65%) than in the other substrate categories.

Dropstones: the presence of dropstones on a photo gives a high probability (95%) of presence of *limpets*, *macroalgae* and *encrusting algae*.

Distance to the glacier: *urchins* and *starfish* have a 100% probability of being absent when the state of the node *distance to the glacier* is “adjacent”. *Laternula* has the highest probability (59%) of being present when “adjacent”, compared to the other states (20% if “close”, 30% if “further”). Similarly, *limpets* have an 81% probability of being present when “adjacent”, 62% when “close” and 28% when “further”. *Macroalgae* are most likely to be abundant when “close” (65%), and in fewer numbers or absent when “adjacent” (66% and 36%, respectively). *Encrusting algae* have a higher probability of being present (54% and 84%) rather than absent when “adjacent” or “close”, and the opposite when “further”. *Suspension feeders* are most likely absent in all cases, with a maximum probability (85%) to be absent when “further” from the glacier. Finally, *dropstones* are more likely to be present (72%) when “adjacent” to the glacier, in fewer numbers when “close”, and absent (67%) when “further”.

Biotic nodes

Starfish: the probability of presence of nodes *laternula* and *urchins* doesn’t change much (<10%) when *starfish* are present or absent. The probability of *limpets* being present is higher when *starfish* are absent (50%) than present (32%). *Macroalgae* are more likely to be present (few: 44%, many: 45%) rather than absent (>1%) when *starfish* are absent. This pattern is similar for *encrusting algae*. *Suspension feeders* are most likely absent in all cases but are more likely to be present when *starfish* are absent (30%) rather than present (>1%).

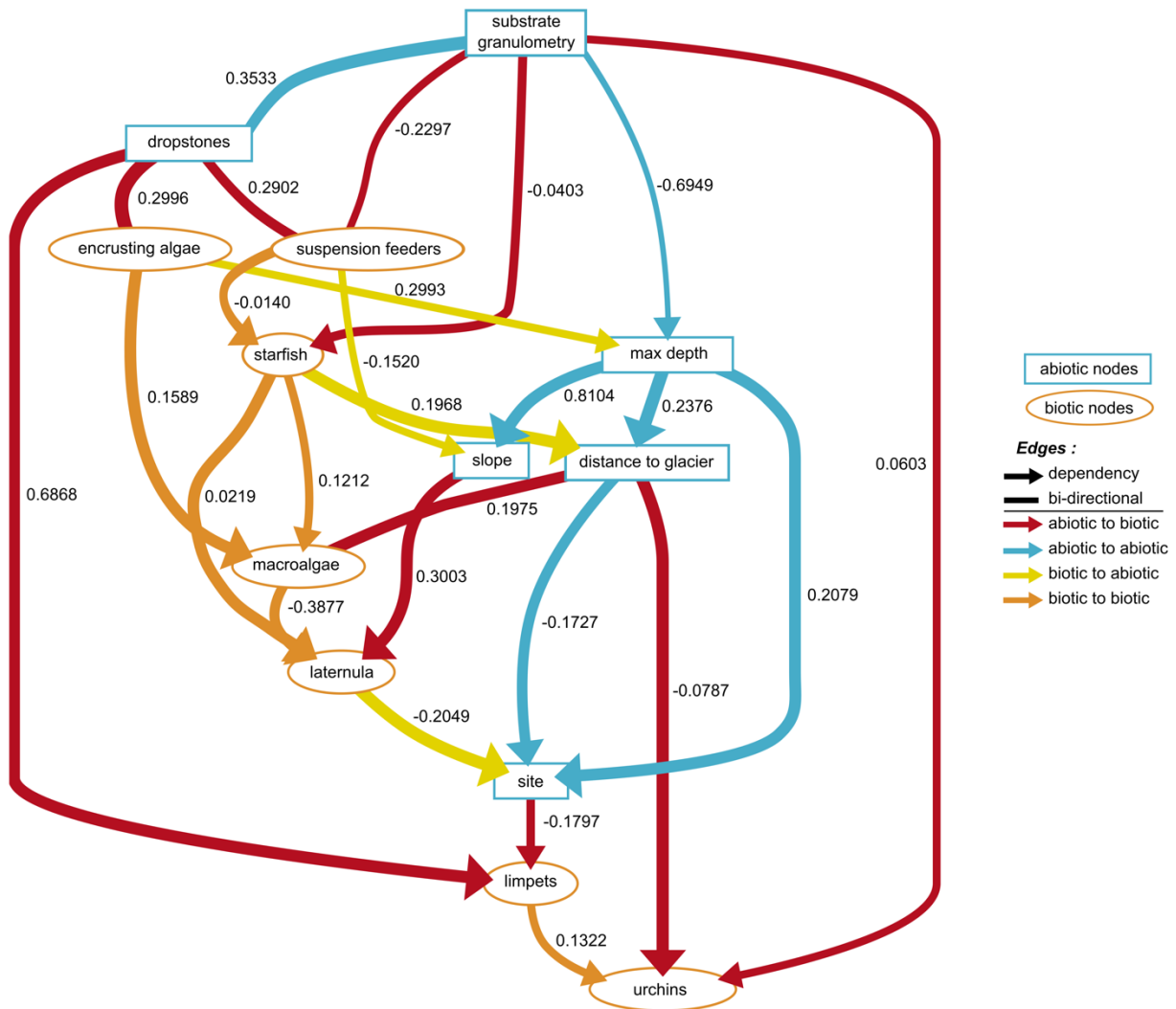


Figure 7 - Bayesian Network of biotic and abiotic factors in Dodman Island. The lines between the nodes represent the edges, or dependencies, between the nodes. The lines are either directed (with an arrowhead in the direction of the dependency) or bi-directional (without an arrowhead). Width of the lines reflects the frequency at which the edge appeared in bootstrapped networks. The colour of the lines represents the type of connection (abiotic to abiotic, abiotic to biotic, etc.). The number next to the line is the Intensity Score (IS) of the connection.

Inferred changes in taxa abundances

In the network, *substrate granulometry* is at the start of every chain (Figure 8). Therefore, if a change occurs in the state of this parameter (i.e. changing from “coarse” to “fine”), we can statistically infer a sequence of effects in the other nodes (Table 3). Given our model, if *substrate granulometry* shifts from “consolidated” to “very fine”, the probability of *starfish* being present increases by 63%, but the probability of presence decreases for *laternula* by 11%, for *urchins* by 44%, for *limpets* by 85%, for *macroalgae* by 22%, for *encrusting algae* by 64%, and for *suspension feeders* by 15%.

Given the numerous relationships involving *starfish* in our network, a change in its state (presence/absence) directly or indirectly impacts to a greater or lesser extent the nodes. Based on our model, if *starfish* shift from “absent” to “present”, *limpets* are 21% more likely to be present, *macroalgae* and *encrusting algae* are 17% less likely to occur and *suspension feeders* are 21% less likely to be present. A shift in *starfish* occurrence doesn’t seem to have a big impact on *laternula* (2% difference) and *urchins* (3% difference).

Since *macroalgae* were abundant over most of our mapped areas, we can infer how the changes in macroalgae abundances impact the other nodes. Indeed, if *macroalgae* go from “absent” to “many”, *laternula* and *starfish* are increasingly likely to be absent (probability increases by 5% and 29% respectively). On the other hand, the probability of presence increases in *urchins* (+9%), *limpets* (+18%), *encrusting algae* (+40%) and *suspension feeders* (+14%).

328 Table 3 – Inferred changes in morphotaxa presence/absence in function of other biotic (if they go from present to absent) and abiotic nodes (if it goes from one state to another, detailed in the name of
329 the column).

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		laternula	urchins	starfish	limpets	macroalgae		encrusting algae	suspension feeders	substrate granulometry			distance to glacier	
						absent- few	few- many			coarse- fine	gravel- fine	rock- fine	adjacent- close	close- far
laternula	Absent	0	-0.04	-0.02	0.07	0.22	-0.27	0.02	0.05	0.09	0.09	0.11	-0.41	0.1
	Present	0	0.06	0.03	-0.1	-0.25	0.28	-0.03	-0.04	-0.1	-0.04	-0.1	0.42	-0.1
urchins	Absent	-0.05	0	-0.03	0.3	0.09	0	0.17	-0.02	0.08	0.35	0.44	0.28	-0.11
	Present	0.04	0	0.02	-0.32	-0.1	-0.01	-0.18	0.01	-0.07	-0.3	-0.42	-0.27	0.12
starfish	Absent	-0.02	0	0	-0.21	-0.29	0	-0.18	-0.36	-0.45	-0.19	-0.63	0.38	0.23
	Present	0.02	0.03	0	0.19	0.3	0	0.19	0.34	0.45	0.17	0.65	-0.4	-0.2
limpets	Absent	0.11	0.5	-0.21	0	0.24	-0.06	0.43	0.2	0.27	0.51	0.83	-0.2	-0.34
	Present	-0.13	-0.51	0.2	0	-0.21	0.07	-0.44	-0.23	-0.27	-0.46	-0.86	0.19	0.34
macroalgae	Absent	0.07	0.11	-0.18	0.12	0	0	0.26	0.14	0.2	0.16	0.19	0.36	-0.22
	Few	-0.32	-0.04	0.08	-0.12	0	0	-0.21	-0.08	-0.04	-0.05	-0.18	0.31	-0.06
encrusting algae	Many	0.23	-0.06	0.09	-0.01	0	0	-0.07	-0.03	-0.18	-0.13	-0.04	-0.67	0.29
	Absent	0.05	0.29	-0.17	0.43	0.47	-0.07	0	0.36	0.38	0.47	0.67	0.31	-0.57
	Present	-0.04	-0.27	0.17	-0.45	-0.47	0.07	0	-0.35	-0.39	-0.5	-0.64	-0.3	0.59
suspension feeders	Absent	0.02	-0.02	-0.21	0.16	0.14	0	0.24	0	0.32	0.04	0.15	0.06	-0.14
	Present	-0.03	0.02	0.23	-0.15	-0.15	0.01	-0.24	0	-0.32	-0.05	-0.16	-0.05	0.15
Mean	Absent	0.04	0.14	0.12	0.18	0.21	0.05	0.18	0.16	0.26	0.26	0.43	0.29	0.24
	Present	0.10	0.12	0.10	0.17	0.18	0.05	0.17	0.13	0.23	0.21	0.38	0.33	0.23

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Discussion

Our results revealed high variability among and within our studied sites, encompassing both taxonomic differences with distinct dominant morphotaxa and functional differences with diverse trophic groups. Sites in intermediate positions—neither too close nor too far from the glacier—showed the highest number of distinct morphotaxa. Site 2 exhibited the highest alpha diversity on Dodman Island, with a more evenly distributed community compared to other sites. While heterogeneity in benthic communities is well-documented in broader-scale studies (Gutt et al. 2013, 2016, 2019; Sahade et al. 2015), our sampling design offers a novel perspective by capturing this variability at a much finer scale.

Using BNI, we identified 26 dependence relationships between both biotic and abiotic factors. **Substrate granulometry** was the focal node of the network, situated at the beginning of every chain, suggesting that this parameter plays a crucial role in the presence of each studied group. Mean inferred changes caused by a change in this node are also the highest compared to all the other nodes (Table 3). Our finding on the importance of substrate granulometry is consistent with findings from other studies around the world (Gallucci et al. 2020; Romoth et al. 2023) and in the Southern Ocean (Post et al. 2017; Brasier et al. 2018; Mitchell et al. 2020). **Distance to the glacier** was also highly connected within the network, which means this parameter influences benthic communities, directly or indirectly (Pasotti et al. 2015; Sahade et al. 2015; Hoffmann et al. 2018; Potocka et al. 2019). Among all our biotic nodes, **starfish** exhibited the most connections. This finding is consistent with the significant role starfish play in shaping communities due to their high abundance in the shallow areas of the Antarctic Peninsula, their broad diet, and their predatory nature (Dearborn et al. 1991; Ha et al. 2019; Moran et al. 2023). Predation relationships are also picked up by our analyses: **starfish** was positively related to **laternula**, a burrowing bivalve (confirmed as *Laternula elliptica*), which is considered one of the prey of *Odontaster validus* (Zamorano et al. 1986), the most abundant starfish in our study area. The **urchins** node comprised only one morphotype, confirmed as *Sterechinus neumayeri*, and appeared at the bottom of all chains, suggesting that its presence can be predicted by examining the state of most other nodes. This network position is consistent with other findings that show that it is a generalist that adapts to most environments (Michel et al. 2016). The BNI also identified the strong connection between **dropstones** and **limpets**, consistent with limpets like *Nacella concinna* that were exclusively seen on dropstones, which is their preferred habitat because of their grazing habits (Suda et al. 2015).

Using our model, we could investigate possible impacts of sea-ice changes in Dodman Island on the benthos. For instance, sea-ice cover in Antarctica will decrease with 15-50% or more by 2100 (Rackow et al. 2022; Holmes et al. 2022), and will cause macroalgae cover to increase (Amsler et al. 2023). Based on our network and inference calculations, we can model the effect of this change would have on other nodes. In our network (Figure 8), the **macroalgae** node is at the start of the following chain: **macroalgae** – **laternula** – **site** – **limpets** – **urchins**. If macroalgae become more abundant, this has a direct effect on **laternula**, which will more likely be absent (+5% probability). We can also infer third- and fourth-degree effects on **limpets** (+18%) and **urchins** (+9%), for which the

probability of presence increases. Therefore, we suggest that if macroalgae becomes more abundant, the impact of this increase will be limited to *laternula*, limpets and urchins, although the scale of the changes is relatively small. Another scenario to explore is the effect of glacier runoff and sedimentation, which is expected to increase with climate change (Siegert et al. 2019), and this increase has been shown to have a direct impact on benthos (Sahade et al. 2015). With our network, we can however also study the indirect effect this has on many other nodes that are downstream. In the following chain: “*substrate granulometry* – *suspension feeders* – *starfish* – *macroalgae* – *laternula*”, if *substrate granulometry* would shift to “fine”, *suspension feeders* are less likely to be present (by 5-32%), *laternula* will behave similarly and will more likely become absent (by 4-10%) and *macroalgae* will more likely be absent (by 16-20%). On the contrary, *starfish* are more likely to be present by 17-65%, which can change the whole dynamic of the community. Although there are many connections between the biotic nodes in our network, changes in their state impact the other nodes less than the abiotic nodes (Table 3). Macroalgae have the highest average impact (20%), but this is still below all the abiotic nodes (Table 3). These results highlight the importance of changing physical and environment on the WAP benthos, over and above the impact that comes from changing species abundances. It is important to note that the inferred Bayesian network is both scale-dependent and time-specific, and its predictive validity is limited to the spatial extent of our study (~1 km²) and the temporal window of sampling (Antarctic summer, February–March).

Conclusion

This study sought to address knowledge gaps concerning small-scale heterogeneity in benthic communities of the WAP and identify the major environmental factors driving these differences. Our findings indicate that within the same depth range, substrate granulometry plays an important role in shaping benthic communities at Dodman Island. High substrate patchiness leads to high benthic heterogeneity, and different granulometry categories define which groups dominate these communities. This conclusion is supported by both our direct observations and BNI results. Our model also indicates that at the scale of a bay such as Dodman Island, an increase in the abundance of macroalgae (like recorded by Amsler et al., (2023) on a larger scale), don’t have as much of an impact on community composition as a change in substrate granulometry or other abiotic factors. Furthermore, we propose a non-destructive, reproducible, cost-effective, and time-efficient method for image acquisition, complemented by a user-friendly workflow that requires minimal expert knowledge. This versatile approach allows our data to be analysed for various applications, including trophic food webs, species distributions, biotic interactions, and spatial patterns. The morphological detail provided by this method is adequate for numerous scientific and ecological applications. We also highlighted the usefulness of combining underwater imagery with BNI, as this approach efficiently identifies real interactions without the need for long-term field observations or extensive datasets. Our BNI modelling can also be used in synergy with other data that predict the effect of climate change, help prioritize sampling efforts, and identify potential cascade effects, which are useful for setting conservation priorities.

Declarations

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Data availability

Raw abundance data and discretized data (for BNI analysis) are both available at:
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Competing Interests

The authors declare no competing interests.

Bibliography

- Althaus F, Hill N, Ferrari R, et al (2015) A Standardised Vocabulary for Identifying Benthic Biota and Substrata from Underwater Imagery: The CATAMI Classification Scheme. PLoS ONE 10:e0141039. <https://doi.org/10.1371/journal.pone.0141039>
- Amsler CD, Amsler MO, Klein AG, et al (2023) Strong correlations of sea ice cover with macroalgal cover along the Antarctic Peninsula: Ramifications for present and future benthic communities. Elem Sci Anth 11:00020. <https://doi.org/10.1525/elementa.2023.00020>
- Arnaubec A, Opderbeke J, Allais AG, Brignone L (2015) Optical mapping with the ARIANE HROV at IFREMER: The MATISSE processing tool. IEEE
- Bellard F (2023) FFmpeg
- Brasier MJ, Grant SM, Trathan PN, et al (2018) Benthic biodiversity in the South Orkney Islands Southern Shelf Marine Protected Area. Biodiversity 1–15. <https://doi.org/10.1080/14888386.2018.1468821>
- Caputi et al. SS (2020) Seasonal Food Web Dynamics in the Antarctic Benthos of Tethys Bay (Ross Sea): Implications for Biodiversity Persistence Under Different Seasonal Sea-Ice Coverage. Front Mar Sci 7:. <https://doi.org/10.3389/fmars.2020.594454>
- Clark GF, Stark JS, Palmer AS, et al (2017) The Roles of Sea-Ice, Light and Sedimentation in Structuring Shallow Antarctic Benthic Communities. PLoS ONE 12:e0168391. <https://doi.org/10.1371/journal.pone.0168391>
- Cummings V, Thrush S, Norkko A, et al (2006) Accounting for local scale variability in benthos: implications for future assessments of latitudinal trends in the coastal Ross Sea. Antarctic science 18:633–644. <https://doi.org/10.1017/S0954102006000666>
- Danis B, Amenabar M, Bombosch A, et al (2023) Report of the TANGO 1 expedition to the West Antarctic Peninsula.
- Danis B, Bayat M, Brusselman A, et al (2024) Report of the TANGO 2 expedition to the West Antarctic Peninsula.
- Dearborn JH, Edwards KC, Fratt DB (1991) Diet, feeding behavior, and surface morphology of the multi-armed Antarctic sea star *Labidiaster annulatus* (Echinodermata: Asteroidea). Marine Ecology Progress Series 77:65–84
- Durden et al. (2016) Perspectives in visual imaging for marine biology and ecology: from acquisition to understanding. Oceanogr Mar Biol Annu Rev 54:1–72. <https://doi.org/doi: 10.5281/zenodo.202685>
- Eayrs C, Li X, Raphael MN, Holland DM (2021) Rapid decline in Antarctic sea ice in recent years hints at future change. Nat Geosci 14:460–464. <https://doi.org/10.1038/s41561-021-00768-3>
- Forio MAE, Moltchanova E, Goethals PLM (2023) Editorial: Application of Bayesian modeling in environmental management. Front Environ Sci 10:1128055. <https://doi.org/10.3389/fenvs.2022.1128055>
- Friedman N, Murphy K, Russell S (1998) Learning the Structure of Dynamic Probabilistic Networks
- Gallucci F, A. Christofolletti R, Fonseca G, M. Dias G (2020) The Effects of Habitat Heterogeneity at Distinct Spatial Scales on Hard-Bottom-Associated Communities. Diversity 12:39. <https://doi.org/10.3390/d12010039>
- Gutt J, Alvaro MC, Barco A, et al (2016) Macroepibenthic communities at the tip of the Antarctic Peninsula, an ecological survey at different spatial scales. Polar Biol 39:829–849. <https://doi.org/10.1007/s00300-015-1797-6>

450 Gutt J, Arndt J, Kraan C, et al (2019) Benthic communities and their drivers: A spatial analysis off the Antarctic
451 Peninsula. *Limnol Oceanogr* 64:2341–2357. <https://doi.org/10.1002/lno.11187>

452 Gutt J, Griffiths HJ, Jones CD (2013) Circumpolar overview and spatial heterogeneity of Antarctic macrobenthic
453 communities. *Mar Biodiv* 43:481–487. <https://doi.org/10.1007/s12526-013-0152-9>

454 Ha S-Y, Ahn I-Y, Moon H-W, et al (2019) Tight trophic association between benthic diatom blooms and shallow-
455 water megabenthic communities in a rapidly deglaciated Antarctic fjord. *Estuarine, Coastal and Shelf Science*
456 218:258–267. <https://doi.org/10.1016/j.ecss.2018.12.020>

457 Heckerman D (1995) Learning Bayesian Networks: The Combination of Knowledge and Statistical Data. *Machine*
458 *Learning* 20:197–243

459 Hewitt (2005) The importance of Small-scale Habitat Structure for maintaining Beta Diversity. *Ecology*

460 Hoffmann R, Pasotti F, Vázquez S, et al (2018) Spatial variability of biogeochemistry in shallow coastal benthic
461 communities of Potter Cove (Antarctica) and the impact of a melting glacier. *PLoS ONE* 13:e0207917.
462 <https://doi.org/10.1371/journal.pone.0207917>

463 Holmes CR, Bracegirdle TJ, Holland PR (2022) Antarctic Sea Ice Projections Constrained by Historical Ice Cover
464 and Future Global Temperature Change. *Geophysical Research Letters* 49:e2021GL097413.
465 <https://doi.org/10.1029/2021GL097413>

466 Horton T, Marsh L, Bett BJ, et al (2021) Recommendations for the Standardisation of Open Taxonomic Nomenclature
467 for Image-Based Identifications. *Front Mar Sci* 8:620702. <https://doi.org/10.3389/fmars.2021.620702>

468 Hui E, Stafford R, Matthews IM, Smith VA (2022) Bayesian networks as a novel tool to enhance interpretability and
469 predictive power of ecological models. *Ecological Informatics* 68:101539.
470 <https://doi.org/10.1016/j.ecoinf.2021.101539>

471 Katz L (2024) CATAMI Label Tree - Western Antarctic Peninsula morphotaxa

472 Kennicutt MC, Chown SL, Cassano JJ, et al (2015) A roadmap for Antarctic and Southern Ocean science for the next
473 two decades and beyond. *Antarctic Science* 27:3–18. <https://doi.org/10.1017/S0954102014000674>

474 Khan TM, Griffiths HJ, Whittle RJ, et al (2024) Network analyses on photographic surveys reveal that invertebrate
475 predators do not structure epibenthos in the deep (~2000m) rocky Powell Basin, Weddell Sea, Antarctica.
476 *Front Mar Sci* 11:1408828. <https://doi.org/10.3389/fmars.2024.1408828>

477 La Mesa M, La Mesa G, Piepenburg D, et al (2022) Spatial patterns and behaviour of notothenioid fishes off the
478 northern Antarctic Peninsula. *Polar Biol* 45:971–985. <https://doi.org/10.1007/s00300-022-03047-y>

479 Langenkämper D, Zurowietz M, Schoening T, Nattkemper TW (2017) BIIGLE 2.0 - Browsing and Annotating Large
480 Marine Image Collections. *Front Mar Sci* 4:. <https://doi.org/10.3389/fmars.2017.00083>

481 Marco Scutari (2010) Learning Bayesian Networks with the bnlearn R Package. *Journal of Statistical Software*
482 35(3):1–22

483 Michel LN, David B, Dubois P, et al (2016) Trophic plasticity of Antarctic echinoids under contrasted environmental
484 conditions. *Polar Biol* 39:913–923. <https://doi.org/10.1007/s00300-015-1873-y>

485 Milns I, Beale CM, Smith VA (2010) Revealing ecological networks using Bayesian network inference algorithms.
486 *Ecology* 91:1892–1899. <https://doi.org/10.1890/09-0731.1>

487 Mindlin J, Shepherd TG, Vera C, Osman M (2021) Combined Effects of Global Warming and Ozone
488 Depletion/Recovery on Southern Hemisphere Atmospheric Circulation and Regional Precipitation.
489 *Geophysical Research Letters* 48:e2021GL092568. <https://doi.org/10.1029/2021GL092568>

490 Mitchell EG, Harris S (2020) Mortality, Population and Community Dynamics of the Glass Sponge Dominated
491 Community “The Forest of the Weird” From the Ridge Seamount, Johnston Atoll, Pacific Ocean. *Front Mar*
492 *Sci* 7:565171. <https://doi.org/10.3389/fmars.2020.565171>

493 Mitchell EG, Wallace MI, Smith VA, et al (2021) Bayesian Network Analysis reveals resilience of the jellyfish *Aurelia*
494 *aurita* to an Irish Sea regime shift. *Sci Rep* 11:3707. <https://doi.org/10.1038/s41598-021-82825-w>

495 Mitchell EG, Whittle RJ, Griffiths HJ (2020) Benthic ecosystem cascade effects in Antarctica using Bayesian network
496 inference. *Communications Biology* 3:. <https://doi.org/10.1038/s42003-020-01310-8>

497 Moran AL, McLachlan RH, Thurber AR (2023) Sea Star Wasting Syndrome Reaches the High Antarctic: Two Recent
498 Outbreaks in McMurdo Sound. *bioRxiv*. <https://doi.org/10.1101/2023.02.17.529025>

499 Oksanen J, Simpson G, Blanchet F, Kindt R, Legendre P, Minchin P, O’Hara R, Solymos P, Stevens M, Szoecs E,
500 Wagner H, Barbour M, Bedward M, Bolker B, Borcard D, Carvalho G, Chirico M, De Caceres M, et al
501 (2022) *vegan: Community Ecology Package*

502 Pasotti F, Manini E, Giovannelli D, et al (2015) Antarctic shallow water benthos in an area of recent rapid glacier
503 retreat. *Marine Ecology* 18

504 Piazza P, Gattone SA, Guzzi A, Schiaparelli S (2020) Towards a robust baseline for long-term monitoring of Antarctic
505 coastal benthos. *Hydrobiologia* 847:1753–1771. <https://doi.org/10.1007/s10750-020-04177-2>

506 Pörtner et al. (2019) IPCC : Special Report on the Ocean and Cryosphere in a Changing Climate (SROC)

507 Post AL, Lavoie C, Domack EW, et al (2017) Environmental drivers of benthic communities and habitat heterogeneity
508 on an East Antarctic shelf. *Antarctic Science* 29:17–32. <https://doi.org/10.1017/S0954102016000468>

509 Potocka M, Kidawa A, Panasiuk A, et al (2019) The Effect of Glacier Recession on Benthic and Pelagic Communities:
510 Case Study in Herve Cove, Antarctica. *JMSE* 7:285. <https://doi.org/10.3390/jmse7090285>

511 Pozsgai G, Baird J, Littlewood NA, et al (2016) Interspecific networks in ground beetle (Coleoptera: Carabidae)
512 assemblages. *Ecological Indicators* 68:134–141. <https://doi.org/10.1016/j.ecolind.2015.11.031>

513 R Core Team (2023) *R: A Language and Environment for Statistical Computing*

514 Rackow T, Danilov S, Goessling HF, et al (2022) Delayed Antarctic sea-ice decline in high-resolution climate change
515 simulations. *Nat Commun* 13:637. <https://doi.org/10.1038/s41467-022-28259-y>

516 Roche DG, O’Dea RE, Kerr KA, et al (2022) Closing the knowledge-action gap in conservation with open science.
517 *Conservation Biology* 36:e13835. <https://doi.org/10.1111/cobi.13835>

518 Romoth K, Darr A, Papenmeier S, et al (2023) Substrate Heterogeneity as a Trigger for Species Diversity in Marine
519 Benthic Assemblages. *Biology* 12:825. <https://doi.org/10.3390/biology12060825>

520 Rossi et al. (2019) Antarctic food web architecture under varying dynamics of sea ice cover. *Scientific Reports*.
521 <https://doi.org/10.1038/s41598-019-48245-7>

522 Sahade R, Lagger C, Torre L, et al (2015) Climate change and glacier retreat drive shifts in an Antarctic benthic
523 ecosystem. *Sci Adv* 1:e1500050. <https://doi.org/10.1126/sciadv.1500050>

524 Sander EL, Wootton JT, Allesina S (2017) Ecological Network Inference From Long-Term Presence-Absence Data.
525 Sci Rep 7:7154. <https://doi.org/10.1038/s41598-017-07009-x>

526 Segelken-Voigt A, Bracher A, Dorschel B, et al (2016) Spatial distribution patterns of ascidians (Ascidacea: Tunicata)
527 on the continental shelves off the northern Antarctic Peninsula. Polar Biol 39:863–879.
528 <https://doi.org/10.1007/s00300-016-1909-y>

529 Siegert M, Atkinson A, Banwell A, et al (2019) The Antarctic Peninsula Under a 1.5°C Global Warming Scenario.
530 Front Environ Sci 7:102. <https://doi.org/10.3389/fenvs.2019.00102>

531 Suda CNK, Vani GS, de Oliveira MF, et al (2015) The biology and ecology of the Antarctic limpet *Nacella concinna*.
532 Polar Biol 38:1949–1969. <https://doi.org/10.1007/s00300-015-1789-6>

533 Turner J, Barrand NE, Bracegirdle TJ, et al (2014) Antarctic climate change and the environment: an update. 23

534 Yu J, Smith VA, Wang PP, et al (2004) Advances to Bayesian network inference for generating causal networks from
535 observational biological data. Bioinformatics 20:3594–3603. <https://doi.org/10.1093/bioinformatics/bth448>

536 Yu J, Smith VA, Wang PP, et al (2002) Using Bayesian Network Inference Algorithms to Recover Molecular Genetic
537 Regulatory Networks

538 Zamorano JH, Duarte WE, Moreno CA (1986) Predation upon *Laternula elliptica* (Bivalvia, Anatinidae): A field
539 manipulation in South Bay, Antarctica. Polar Biol 6:139–143. <https://doi.org/10.1007/BF00274876>

540